

TRITACE®  2.5

TRITACE®  5

TRITACE®  10

Active ingredient: Ramipril

QUALITATIVE AND QUANTITATIVE COMPOSITION

Tritace 2.5mg tablets: One tablet contains 2,5 mg ramipril.

Tritace 5 mg tablets: One tablet contains 5 mg ramipril.

Tritace 10 mg tablets: One tablet contains 10 mg ramipril.

For a full list of excipients see *List of Excipients*.

PHARMACEUTICAL FORM

Tritace® 2.5: Yellowish to yellow oblong tablets with score-line,
Upper stamp: 2.5 and company logo, Lower stamp: HMR and 2.5

Tritace® 5: Pale red oblong tablets with score-line,
Upper stamp: 5 and company logo, Lower stamp: HMP and 5

Tritace® 10: White to almost white oblong tablets with score-line,
Upper stamp: HMO/HMO, Lower stamp: anonymous

CLINICAL PARTICULARS

Therapeutic indications

- Hypertension
- Congestive heart failure
- Treatment of patients who – within the first few days after an acute myocardial infarction – have demonstrated clinical signs of congestive heart failure
- For reducing the risk of myocardial infarction, stroke, cardiovascular death or the need for revascularization procedures in patients 55 years or more who have clinical evidence of coronary artery disease, stroke, or peripheral vascular disease.
- For reducing the risk of myocardial infarction, stroke, cardiovascular death or revascularisation procedures in diabetic patients 55 years or more with one or more of the following risk factors: systolic blood pressure > 160 mmHg or diastolic blood pressure > 90 mmHg (or on antihypertensive treatment); total cholesterol > 5.2 mmol/L; HDL cholesterol < 0.9 mmol/L; current smoker; known microalbuminuria; any evidence of previous vascular disease.
- Prevention of progressive renal failure in patients with persistent proteinuria.

Posology and method of administration

Oral use.

It is recommended that Tritace is taken each day at the same time of the day.

Can be taken before, with or after meals, because food intake does not modify its bioavailability (see *Pharmacokinetic properties*). The tablet has to be swallowed with liquid. It must not be chewed or crushed.

Dosage

The dosage is based on the desired effect and on how the patient tolerates the medicine. Therapy with Tritace is usually long-term therapy; the doctor determines the duration of treatment individually for each patient.

Treatment of hypertension

Tritace may be used in monotherapy or in combination with other classes of antihypertensive medicinal products.

The recommended initial dose is 2.5 mg once daily. Depending on the response, the dose may be increased. Any increase should be implemented by doubling the dose at intervals of 2 to 4 weeks. The usual maintenance dose is 2.5 to 5 mg daily, the maximum dose is 10 mg daily.

In patients with a strongly activated renin-angiotensin-aldosterone system, incompletely corrected fluid or salt deficiency, those with severe hypertension, as well as in those for whom a hypotensive reaction would constitute a particular risk (e.g. patients with haemodynamically relevant stenoses of the coronary arteries or of the blood vessels supplying the brain) and in the elderly, a reduced initial dose of 1.25 mg daily must be considered.

Diuretic-Treated patients

In patients pre-treated with a diuretic, consideration must be given to discontinuing the diuretic for at least 2 to 3 days or – depending on the duration of action of the diuretic – longer before starting treatment with Tritace, or at least to reducing the diuretic dose. The doctor will decide in each individual case whether such discontinuation or dose reduction is possible and how long it should last.

In hypertensive patients in whom the diuretic is not discontinued, therapy with Tritace should be initiated with a 1.25 mg dose. Renal function and serum potassium should be monitored. The subsequent dosage of Tritace should be adjusted according to blood pressure target.

Treatment of congestive heart failure

The recommended initial dose is 1.25 mg once daily. Depending on the response, the dose may be increased. Any increase should be implemented by doubling the dose at intervals of 1 to 2 weeks. The maximum daily dose is 10 mg. The required daily dose, if equaling or exceeding 2.5 mg, may be taken as a single dose or in two separate doses (preferable).

Secondary prevention after acute myocardial infarction and with heart failure

The recommended initial dose is 5 mg daily, divided into two single doses of 2.5 mg each, one in the morning and one in the evening for three days. If this dose is not well tolerated, 1.25 mg should be taken twice daily over two days. In either event, depending on the response, the dose may then be increased. Any increase should be implemented by doubling the dose at intervals of 1 to 3 days to the target maintenance dose of 5 mg twice daily. The maintenance dose is divided in 2 administration per day where possible. If the dose cannot be increased to 2.5mg twice a day treatment should be withdrawn.

Sufficient experience is still lacking in the treatment of patients with severe (NYHA IV) heart failure immediately after myocardial infarction. Treatment, if nevertheless given, should be started with 1.25 mg once daily, and increased only with particular caution.

Prevention of progressive renal failure in patients with persistent proteinuria

The recommended initial dose is 1.25mg Tritace once daily. This should be doubled at intervals of 2 weeks, depending on how the drug is tolerated. There are no efficacy data regarding doses above 5mg/day in patients with nephropathy.

In hypertensive patients, a target diastolic blood pressure of <90mmHg should be pursued. In patients pre-treated with a diuretic, consideration must be given to discontinuing the diuretic for at least 2-3 days or longer (depending on duration of action) or at least consideration should be given to reducing the dose, before initiating Tritace.

Dosage in patients at increased cardiovascular risk/ diabetes and at least one CV risk

The recommended initial dose is 2.5 mg once daily. Depending on the tolerability, the dose should be doubled after one or two week of treatment and, after another two to three weeks, should be increased to 10 mg. The usual maintenance dose is Tritace 10 mg daily.

See also posology on diuretic treated patients above.

Special populations

Dosage in patients with impaired renal function

Daily dose in patients with renal impairment should be based on creatinine clearance (see *Pharmacokinetic properties*)

- if creatinine clearance is ≥ 60 ml/min, it is not necessary to adjust the initial dose (2.5mg/day); the maximal daily dose is 10mg- if creatinine clearance is between 30-60 ml/min, it is not necessary to adjust the initial dose (2.5 mg/day); the maximal daily dose is 5 mg;
- if creatinine clearance is between 10-30 ml/min, the initial dose is 1.25 mg/day and the maximal daily dose is 5 mg;
- in haemodialysed hypertensive patients: ramipril is slightly dialysable; the initial dose is 1.25 mg/day and the maximal daily dose is 5 mg; the medicinal product should be administered few hours after haemodialysis is performed.

Particular care should be exercised in patients with impaired renal function who are to be treated for heart failure post MI as such patients may be susceptible to hypotension. Patients with impaired renal function treated for heart failure post MI have not been studied systematically.

Dosage in patients with impaired liver function

In patients with impaired liver function, the metabolism of ramipril – and therefore the formation of the bioactive metabolite ramiprilat – is delayed due to diminished activity of the esterases in the liver, resulting in elevated plasma ramipril levels. Treatment with Tritace should therefore be initiated under close medical supervision and should not exceed 2.5 mg daily.

Dosage in elderly

The recommended starting dose is 1.25 mg once daily, which can then be increased according to the individual patient's BP response.

Paediatric population

Tritace is not recommended for use in children and adolescents below 18 years of age due to insufficient data on safety and efficacy.

Contraindications

- Hypersensitivity to the active substance, to any of the excipients or any other ACE (Angiotensin Converting Enzyme) inhibitors (see *List of excipients*)

- History of angioedema (hereditary, idiopathic or due to previous angioedema with ACE inhibitors or AIIRAs)
- Concomitant use with sacubitril/valsartan therapy (see sections *Special warnings and precautions for use* and *Interaction with other medicinal products and other forms of interaction*)
- Extracorporeal treatments leading to contact of blood with negatively charged surfaces (see *Interaction with other medicinal products and other forms of interaction*)
- Significant bilateral renal artery stenosis or renal artery stenosis in a single functioning kidney
- 2nd and 3rd trimester of pregnancy (see *Special warnings and precautions for use* and *Pregnancy and lactation*)
- Ramipril must not be used in patients with hypotensive or haemodynamically unstable states.
- The concomitant use of Tritace with aliskiren-containing products is contraindicated in patients with diabetes mellitus or renal impairment (GFR < 60 ml/min/1.73m²)(see *Interaction with other medicinal products and other forms of interaction* and *Pharmacodynamic properties*)

Special warnings and precautions for use

Special populations

- *Pregnancy*: ACE inhibitors such as ramipril, or Angiotensin II Receptor Antagonists (AIIRAs) should not be initiated during pregnancy. Unless continued ACE inhibitor/ AIIRAs therapy is considered essential, patients planning pregnancy should be changed to alternative anti-hypertensive treatments which have an established safety profile for use in pregnancy. When pregnancy is diagnosed, treatment with ACE inhibitors/ AIIRAs should be stopped immediately, and, if appropriate, alternative therapy should be started (see *Contraindications* and *Pregnancy and lactation*).

- *Patients at particular risk of hypotension*

- *Patients with strongly activated renin-angiotensin-aldosterone system*

Patients with strongly activated renin-angiotensin-aldosterone system are at risk of an acute pronounced fall in blood pressure and deterioration of renal function due to ACE inhibition, especially when an ACE inhibitor or a concomitant diuretic is given for the first time or at first dose increase.

Significant activation of renin-angiotensin-aldosterone system is to be anticipated and medical supervision including blood pressure monitoring is necessary, for example in:

- patients with severe hypertension
- patients with decompensated congestive heart failure
- patients with haemodynamically relevant left ventricular inflow or outflow impediment (e.g. stenosis of the aortic or mitral valve)
- patients with unilateral renal artery stenosis with a second functional kidney
- patients in whom fluid or salt depletion exists or may develop (including patients with diuretics)
- patients with liver cirrhosis and/or ascites
- patients undergoing major surgery or during anaesthesia with agents that produce hypotension.

Generally, it is recommended to correct dehydration, hypovolaemia or salt depletion before initiating treatment (in patients with heart failure, however, such corrective action must be carefully weighed out against the risk of volume overload).

- *Dual blockade of the renin-angiotensin-aldosterone system (RAAS)*

There is evidence that the concomitant use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren increases the risk of hypotension, hyperkalaemia and decreased renal function

(including acute renal failure). Dual blockade of RAAS through the combined use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren is therefore not recommended (see *Interaction with other medicinal products and other forms of interaction* and *Pharmacodynamic properties*).

If dual blockade therapy is considered absolutely necessary, this should only occur under specialist supervision and subject to frequent close monitoring of renal function, electrolytes and blood pressure.

ACE-inhibitors and angiotensin II receptor blockers should not be used concomitantly in patients with diabetic nephropathy.

- *Transient or persistent heart failure post MI*

- *Patients at risk of cardiac or cerebral ischemia in case of acute hypotension*

The initial phase of treatment requires special medical supervision.

- *Elderly patients*

See *Posology and method of administration*.

Surgery

It is recommended that treatment with angiotensin converting enzyme inhibitors such as ramipril should be discontinued where possible one day before surgery.

Monitoring of renal function

Renal function should be assessed before and during treatment and dosage adjusted especially in the initial weeks of treatment. Particularly careful monitoring is required in patients with renal impairment (see *Posology and method of administration*). There is a risk of impairment of renal function, particularly in patients with congestive heart failure or after a renal transplant.

Angioedema

Angioedema has been reported in patients treated with ACE inhibitors including ramipril (see *Undesirable effects*). This risk of angioedema may be increased in patients taking concomitant medications which may cause angioedema such as mTOR (mammalian target of rapamycin) inhibitors (e.g. temsirolimus, everolimus, sirolimus), vildagliptin or neprilysin (NEP) inhibitors (such as racecadotril). The combination of ramipril with sacubitril/valsartan is contraindicated due to the increased risk of angioedema (see sections *Contraindications* and *Interaction with other medicinal products and other forms of interaction*).

In case of angioedema, Tritace must be discontinued.

Emergency therapy should be instituted promptly. Patient should be kept under observation for at least 12 to 24 hours and discharged after complete resolution of the symptoms.

Intestinal angioedema has been reported in patients treated with ACE inhibitors including Tritace (see *Undesirable effects*). These patients presented with abdominal pain (with or without nausea or vomiting).

Anaphylactic reactions during desensitization

The likelihood and severity of anaphylactic and anaphylactoid reactions to insect venom and other allergens are increased under ACE inhibition. A temporary discontinuation of Tritace should be considered prior to desensitization.

Electrolyte Monitoring: Hyperkalaemia

Hyperkalaemia has been observed in some patients treated with ACE inhibitors including Tritace. Patients at risk for development of hyperkalaemia include those with renal insufficiency, age (> 70 years), uncontrolled diabetes mellitus, or those using potassium salts, potassium retaining diuretics and other plasma potassium increasing active substances, or

conditions such as dehydration, acute cardiac decompensation, metabolic acidosis. If concomitant use of the above mentioned agents is deemed appropriate, regular monitoring of serum potassium is recommended (see *Interaction with other medicinal products and other forms of interaction*).

Electrolyte Monitoring: Hyponatraemia

Syndrome of Inappropriate Anti-diuretic Hormone (SIADH) and subsequent hyponatraemia has been observed in some patients treated with ramipril. It is recommended that serum sodium levels be monitored regularly in the elderly and in other patients at risk of hyponatraemia.

Neutropenia/agranulocytosis

Neutropenia/agranulocytosis, as well as thrombocytopenia and anaemia, have been rarely seen and bone marrow depression has also been reported. It is recommended to monitor the white blood cell count to permit detection of a possible leucopenia. More frequent monitoring is advised in the initial phase of treatment and in patients with impaired renal function, those with concomitant collagen disease (e.g. lupus erythematosus or scleroderma), and all those treated with other medicinal products that can cause changes in the blood picture (see *Interaction with other medicinal products and other forms of interaction* and Undesirable effects).

Ethnic differences

ACE inhibitors cause higher rate of angioedema in black patients than in non black patients. As with other ACE inhibitors, ramipril may be less effective in lowering blood pressure in black people than in non black patients, possibly because of a higher prevalence of hypertension with low renin level in the black hypertensive population.

Cough

Cough has been reported with the use of ACE inhibitors. Characteristically, the cough is non-productive, persistent and resolves after discontinuation of therapy. ACE inhibitor-induced cough should be considered as part of the differential diagnosis of cough.

Effects on ability to drive and use machines

Some adverse effects (e.g. symptoms of a reduction in blood pressure such as dizziness) may impair the patient's ability to concentrate and react and, therefore, constitute a risk in situations where these abilities are of particular importance (e.g. operating a vehicle or machinery). This can happen especially at the start of treatment, or when changing over from other preparations. After the first dose or subsequent increases in dose it is not advisable to drive or operate machinery for several hours.

Interaction with other medicinal products and other forms of interaction

Clinical trial data has shown that dual blockade of the renin-angiotensin-aldosterone-system (RAAS) through the combined use of ACE-inhibitors, angiotensin II receptor blockers or aliskiren is associated with a higher frequency of adverse events such as hypotension, hyperkalaemia and decreased renal function (including acute renal failure) compared to the use of a single RAAS-acting agent (see *Contraindications, Special warnings and precautions for use* and *Pharmacodynamic properties*).

Contraindicated combinations

The concomitant use of ACE inhibitors with sacubitril/valsartan is contraindicated as this increases the risk of angioedema (see sections *Contraindications* and *Special warnings and precautions for use*). Treatment with ramipril must not be started until 36 hours after taking the last dose of sacubitril/valsartan. Sacubitril/valsartan must not be started until 36 hours after the last dose of TRITACE.

Extracorporeal treatments leading to contact of blood with negatively charged surfaces such as dialysis or haemofiltration with certain high-flux membranes (e.g. polyacrylonitril membranes) and low density lipoprotein apheresis with dextran sulphate due to increased risk of severe anaphylactoid reactions (see *Contraindications*). If such treatment is required, consideration should be given to using a different type of dialysis membrane or a different class of antihypertensive agent.

Precautions for use

Potassium salts, heparin, potassium-retaining diuretics and other plasma potassium increasing active substances (including Angiotensin II antagonists, trimethoprim and in fixed dose combination with sulfamethoxazole, tacrolimus, ciclosporin): Hyperkalaemia may occur, therefore close monitoring of serum potassium is required.

Antihypertensive agents (e.g. diuretics) and other substances that may decrease blood pressure (e.g. nitrates, tricyclic antidepressants, anaesthetics, acute alcohol intake, baclofen, alfuzosin, doxazosin, prazosin, tamsulosin, terazosin): Potentiation of the risk of hypotension is to be anticipated (see *Posology and method of administration for diuretics*)

Vasopressor sympathomimetics and other substances (e.g. isoproterenol, dobutamine, dopamine, epinephrine) that may reduce the antihypertensive effect of Tritace: Blood pressure monitoring is recommended.

Allopurinol, immunosuppressants, corticosteroids, procainamide, cytostatics and other substances that may change the blood cell count: Increased likelihood of haematological reactions (see *Special warnings and precautions for use*).

Lithium salts: Excretion of lithium may be reduced by ACE inhibitors and therefore lithium toxicity may be increased. Lithium level must be monitored.

Antidiabetic agents including insulin: Hypoglycaemic reactions may occur. Blood glucose monitoring is recommended.

Non-steroidal anti-inflammatory drugs and acetylsalicylic acid: Reduction of the antihypertensive effect of Tritace is to be anticipated. Furthermore, concomitant treatment of ACE inhibitors and NSAIDs may lead to an increased risk of worsening of renal function and to an increase in kalaemia.

mTOR inhibitors or DPP-IV inhibitors: An increased risk of angioedema is possible in patients taking concomitant medications such as mTOR inhibitors (e.g. temsirolimus, everolimus, sirolimus) or vildagliptin. Caution should be used when starting therapy (see *Special warnings and precautions for use*).

Neprilysin (NEP) inhibitors: An increased risk of angioedema has been reported with concomitant use of ACE inhibitors and NEP inhibitor such as racecadotril (see section *Special warnings and precautions for use*).

Sacubitril/valsartan: The concomitant use of ACE inhibitors with sacubitril/valsartan is contraindicated as this increases the risk of angioedema.

Fertility, pregnancy and lactation

Pregnancy

Tritace is not recommended during the first trimester of pregnancy (see *Special warnings and precautions for use*) and contraindicated during the second and third trimesters of pregnancy (see *Contraindications*).

Epidemiological evidence regarding the risk of teratogenicity following exposure to ACE inhibitors during the first trimester of pregnancy has not been conclusive; however a small increase in risk cannot be excluded. Unless continued ACE inhibitor therapy is considered essential, patients planning pregnancy should be changed to alternative anti-hypertensive treatments which have an established safety profile for use in pregnancy. When pregnancy is diagnosed, treatment with ACE inhibitors should be stopped immediately, and, if appropriate, alternative therapy should be started.

ACE inhibitor/ Angiotensin II Receptor Antagonist (AIIRA) therapy exposure during the second and third trimesters is known to induce human fetotoxicity (decreased renal function, oligohydramnios, skull ossification retardation) and neonatal toxicity (renal failure, hypotension, hyperkalaemia) (See *Preclinical safety data*). Should exposure to ACE inhibitor have occurred from the second trimester of pregnancy, ultrasound check of renal function and skull is recommended. Newborns whose mothers have taken ACE inhibitors should be closely observed for hypotension, oliguria and hyperkalaemia (see *Contraindications* and *Special warnings and precautions for use*).

Breastfeeding

Because insufficient information is available regarding the use of ramipril during breastfeeding (see *Pharmacokinetic properties*), ramipril is not recommended and alternative treatments with better established safety profiles during breast-feeding are preferable, especially while nursing a newborn or preterm infant.

Undesirable effects

Summary of safety profile

The safety profile of ramipril includes persistent dry cough and reactions due to hypotension. Serious adverse reactions include angioedema, hyperkalaemia, renal or hepatic impairment, pancreatitis, severe skin reactions and neutropenia/agranulocytosis.

Tabulated list of adverse reactions

Adverse reactions frequency is defined using the following convention:

Very common ($\geq 1/10$); common ($\geq 1/100$ to $< 1/10$); uncommon ($\geq 1/1,000$ to $< 1/100$); rare ($\geq 1/10,000$ to $< 1/1,000$); very rare ($< 1/10,000$), not known (cannot be estimated from the available data).

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

	Common	Uncommon	Rare	Very rare	Not known
<u>Cardiac disorders</u>		Myocardial ischaemia including angina pectoris or myocardial infarction, tachycardia, arrhythmia, palpitations, oedema peripheral			

<u>Blood and lymphatic system disorders</u>		Eosinophilia	White blood cell count decreased (including neutropenia or agranulocytosis), red blood cell count decreased, haemoglobin decreased, platelet count decreased		Bone marrow failure, pancytopenia, haemolytic anaemia
<u>Nervous system disorders</u>	Headache, dizziness	Vertigo, paraesthesia, ageusia, dysgeusia,	Tremor, balance disorder		Cerebral ischaemia including ischaemic stroke and transient ischaemic attack, psychomotor skills impaired, burning sensation, parosmia
<u>Endocrine disorders</u>					Syndrome of inappropriate antidiuretic hormone secretion (SIADH)
<u>Eye disorders</u>		Visual disturbance including blurred vision	Conjunctivitis		
<u>Ear and labyrinth disorders</u>			Hearing impaired, tinnitus		
<u>Respiratory, thoracic and mediastinal disorders</u>	Non-productive tickling cough, bronchitis, sinusitis, dyspnoea	Bronchospasm including asthma aggravated, nasal congestion			
<u>Gastrointestinal disorders</u>	Gastrointestinal inflammation, digestive disturbances, abdominal discomfort, dyspepsia,	Pancreatitis (cases of fatal outcome have been very exceptionally reported with ACE inhibitors),	Glossitis		Aphtous stomatitis

	diarrhoea, nausea, vomiting	pancreatic enzymes increased, small bowel angioedema, abdominal pain upper including gastritis, constipation, dry mouth			
<u>Renal and urinary disorders</u>		Renal impairment including renal failure acute, urine output increased, worsening of a pre-existing proteinuria, blood urea increased, blood creatinine increased			
<u>Skin and subcutaneous tissue disorders</u>	Rash in particular maculo- papular	Angioedema; very exceptionally, the airway obstruction resulting from angioedema may have a fatal outcome; pruritus, hyperhidrosis	Exfoliative dermatitis, urticaria, onycholysis,	Photosensitivity reaction	Toxic epidermal necrolysis, Stevens- Johnson syndrome, erythema multiforme, pemphigus, psoriasis aggravated, dermatitis psoriasiform, pemphigoid or lichenoid exanthema or enanthema, alopecia
<u>Musculoskeletal and connective tissue disorders</u>	Muscle spasms, myalgia	Arthralgia			
<u>Metabolism and nutrition disorders</u>	Blood potassium increased	Anorexia, decreased appetite,			Blood sodium decreased

<u>Vascular disorders</u>	Hypotension, orthostatic blood pressure decreased, syncope	Flushing	Vascular stenosis, hypoperfusion, vasculitis		Raynaud's phenomenon
<u>General disorders and administration site conditions</u>	Chest pain, fatigue	Pyrexia	Asthenia		
<u>Immune system disorders</u>					Anaphylactic or anaphylactoid reactions, antinuclear antibody increased
<u>Hepatobiliary disorders</u>		Hepatic enzymes and/or bilirubin conjugated increased,	Jaundice cholestatic, hepatocellular damage		Acute hepatic failure, cholestatic or cytolytic hepatitis (fatal outcome has been very exceptional).
<u>Reproductive system and breast disorders</u>		Transient erectile impotence, libido decreased			Gynaecomastia
<u>Psychiatric disorders</u>		Depressed mood, anxiety, nervousness, restlessness, sleep disorder including somnolence	Confusional state		Disturbance in attention

Overdose

Symptoms associated with overdose of ACE inhibitors may include excessive peripheral vasodilation (with marked hypotension, shock), bradycardia, electrolyte disturbances, and renal failure. The patient should be closely monitored and the treatment should be symptomatic and supportive. Suggested measures include primary detoxification (gastric lavage, administration of adsorbents) and measures to restore haemodynamic stability, including, administration of alpha 1 adrenergic agonists or angiotensin II (angiotensinamide) administration. Ramiprilat, the active metabolite of ramipril is poorly removed from the general circulation by haemodialysis.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: ACE Inhibitors, plain, ATC code C09AA05.

Mechanism of action

Ramiprilat, the active metabolite of the prodrug ramipril, inhibits the enzyme dipeptidylcarboxypeptidase I (synonyms: angiotensin-converting enzyme; kininase II). In plasma and tissue this enzyme catalyses the conversion of angiotensin I to the active vasoconstrictor substance angiotensin II, as well as the breakdown of the active vasodilator bradykinin. Reduced angiotensin II formation and inhibition of bradykinin breakdown lead to vasodilatation.

Since angiotensin II also stimulates the release of aldosterone, ramiprilat causes a reduction in aldosterone secretion. The average response to ACE inhibitor monotherapy was lower in black (Afro-Caribbean) hypertensive patients (usually a low-renin hypertensive population) than in non-black patients.

Pharmacodynamic effects

Antihypertensive properties:

Administration of ramipril causes a marked reduction in peripheral arterial resistance. Generally, there are no major changes in renal plasma flow and glomerular filtration rate. Administration of ramipril to patients with hypertension leads to a reduction in supine and standing blood pressure without a compensatory rise in heart rate.

In most patients the onset of the antihypertensive effect of a single dose becomes apparent 1 to 2 hours after oral administration. The peak effect of a single dose is usually reached 3 to 6 hours after oral administration. The antihypertensive effect of a single dose usually lasts for 24 hours.

The maximum antihypertensive effect of continued treatment with ramipril is generally apparent after 3 to 4 weeks. It has been shown that the antihypertensive effect is sustained under long term therapy lasting 2 years.

Abrupt discontinuation of ramipril does not produce a rapid and excessive rebound increase in blood pressure.

Heart failure:

In addition to conventional therapy with diuretics and optional cardiac glycosides, ramipril has been shown to be effective in patients with functional classes II-IV of the New-York Heart Association. The drug had beneficial effects on cardiac haemodynamics (decreased left and right ventricular filling pressures, reduced total peripheral vascular resistance, increased cardiac output and improved cardiac index). It also reduced neuroendocrine activation.

Clinical efficacy and safety

Cardiovascular prevention/Nephroprotection:

A preventive placebo-controlled study (the HOPE-study), was carried out in which ramipril was added to standard therapy in more than 9,200 patients. Patients with increased risk of cardiovascular disease following either atherothrombotic cardiovascular disease (history of coronary heart disease, stroke or peripheral vascular disease) or diabetes mellitus with at least one additional risk factor (documented microalbuminuria, hypertension, elevated total cholesterol level, low high-density lipoprotein cholesterol level or cigarette smoking) were included in the study.

The study showed that ramipril statistically significantly decreases the incidence of myocardial infarction, death from cardiovascular causes and stroke, alone and combined (primary combined events).

The HOPE study: Main results:

	Ramipril	Placebo	relative risk	p-value
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			(95% confidence interval)	
	%	%		
All patients	n=4,645	N=4,652		
Primary combined events	14.0	17.8	0.78 (0.70-0.86)	<0.001
<i>Myocardial infarction</i>	9.9	12.3	0.80 (0.70-0.90)	<0.001
<i>Death from cardiovascular causes</i>	6.1	8.1	0.74 (0.64-0.87)	<0.001
<i>Stroke</i>	3.4	4.9	0.68 (0.56-0.84)	<0.001
Secondary endpoints				
<i>Death from any cause</i>	10.4	12.2	0.84 (0.75-0.95)	0.005
<i>Need for Revascularisation</i>	16.0	18.3	0.85 (0.77-0.94)	0.002
<i>Hospitalisation for unstable angina</i>	12.1	12.3	0.98 (0.87-1.10)	NS
<i>Hospitalisation for heart failure</i>	3.2	3.5	0.88 (0.70-1.10)	0.25
<i>Complications related to diabetes</i>	6.4	7.6	0.84 (0.72-0.98)	0.03

The MICRO-HOPE study, a predefined substudy from HOPE, investigated the effect of the addition of ramipril 10 mg to the current medical regimen versus placebo in 3,577 patients at least ≥ 55 years old (with no upper limit of age), with a majority of type 2 diabetes (and at least another CV risk factor), normotensive or hypertensive.

The primary analysis showed that 117 (6.5 %) participants on ramipril and 149 (8.4 %) on placebo developed overt nephropathy, which corresponds to a RRR 24 %; 95 % CI [3-40], $p = 0.027$.

The REIN study, a multicenter randomized, double-blind parallel group, placebo-controlled study aimed at assessing the effect of treatment with ramipril on the rate of decline of glomerular function rate (GFR) in 352 normotensive or hypertensive patients (18-70 years old) suffering from mild (i.e. mean urinary protein excretion > 1 and < 3 g/24 h) or severe proteinuria (≥ 3 g/24 h) due to chronic non-diabetic nephropathy. Both subpopulations were prospectively stratified.

The main analysis of patients with the most severe proteinuria (stratum prematurely disrupted due to benefit in ramipril group) showed that the mean rate of GFR decline per month was lower with ramipril than with placebo; -0.54 (0.66) vs. -0.88 (1.03) ml/min/month, $p = 0.038$. The intergroup difference was thus 0.34 [0.03-0.65] per month, and around 4 ml/min/year; 23.1 % of the patients in the ramipril group reached the combined secondary endpoint of doubling of baseline serum creatinine concentration and/or end-stage renal disease (ESRD) (need for dialysis or renal transplantation) vs. 45.5 % in the placebo group ($p = 0.02$).

Dual blockade of the renin-angiotensin-aldosterone system (RAAS):

Two large randomised, controlled trials (ONTARGET (ONgoing Telmisartan Alone and in combination with Ramipril Global Endpoint Trial) and VA NEPHRON-D (The Veterans Affairs Nephropathy in Diabetes)) have examined the use of the combination of an ACE-inhibitor with an angiotensin II receptor blocker.

ONTARGET was a study conducted in patients with a history of cardiovascular or cerebrovascular disease, or type 2 diabetes mellitus accompanied by evidence of end-organ damage. VA NEPHRON-D was a study in patients with type 2 diabetes mellitus and diabetic nephropathy.

These studies have shown no significant beneficial effect on renal and/or cardiovascular outcomes and mortality, while an increased risk of hyperkalaemia, acute kidney injury and/or hypotension as compared to monotherapy was observed. Given their similar pharmacodynamic properties, these results are also relevant for other ACE-inhibitors and angiotensin II receptor blockers.

ACE-inhibitors and angiotensin II receptor blockers should therefore not be used concomitantly in patients with diabetic nephropathy.

ALTITUDE (Aliskiren Trial in Type 2 Diabetes Using Cardiovascular and Renal Disease Endpoints) was a study designed to test the benefit of adding aliskiren to a standard therapy of an ACE-inhibitor or an angiotensin II receptor blocker in patients with type 2 diabetes mellitus and chronic kidney disease, cardiovascular disease, or both. The study was terminated early because of an increased risk of adverse outcomes. Cardiovascular death and stroke were both numerically more frequent in the aliskiren group than in the placebo group and adverse events and serious adverse events of interest (hyperkalaemia, hypotension and renal dysfunction) were more frequently reported in the aliskiren group than in the placebo group.

Secondary prevention after acute myocardial infarction

The AIRE study included more than 2,000 patients with transient/persistent clinical signs of heart failure after documented myocardial infarction. The ramipril treatment was started 3 to 10 days after the acute myocardial infarction. The study showed that after an average follow-up time of 15 months the mortality in ramipril-treated patients was 16.9 % and in the placebo treated patients was 22.6 %. This means an absolute mortality reduction of 5.7 % and a relative risk reduction of 27 % (95 % CI [11-40 %]).

Pharmacokinetic properties

Pharmacokinetics and Metabolism

Absorption

Following oral administration ramipril is rapidly absorbed from the gastrointestinal tract: peak plasma concentrations of ramipril are reached within one hour. Based on urinary recovery, the extent of absorption is at least 56 % and is not significantly influenced by the presence of food in the gastrointestinal tract. The bioavailability of the active metabolite ramiprilat after oral administration of 2.5 mg and 5 mg ramipril is 45 %.

Peak plasma concentrations of ramiprilat, the sole active metabolite of ramipril are reached 2-4 hours after ramipril intake. Steady state plasma concentrations of ramiprilat after once daily dosing with the usual doses of ramipril are reached by about the fourth day of treatment.

Distribution

The serum protein binding of ramipril is about 73 % and that of ramiprilat about 56 %.

Biotransformation

Ramipril is almost completely metabolised to ramiprilat, and to the diketopiperazine ester, the diketopiperazine acid, and the glucuronides of ramipril and ramiprilat.

Elimination

Excretion of the metabolites is primarily renal.

Plasma concentrations of ramiprilat decline in a polyphasic manner. Because of its potent, saturable binding to ACE and slow dissociation from the enzyme, ramiprilat shows a prolonged terminal elimination phase at very low plasma concentrations.

After multiple once-daily doses of ramipril, the effective half-life of ramiprilat concentrations was 13-17 hours for the 5-10 mg doses and longer for the lower 1.25-2.5 mg doses. This difference is related to the saturable capacity of the enzyme to bind ramiprilat.

Patients with renal impairment (see *Posology and method of administration*)

Renal excretion of ramiprilat is reduced in patients with impaired renal function, and renal ramiprilat clearance is proportionally related to creatinine clearance. This results in elevated plasma concentrations of ramiprilat, which decrease more slowly than in subjects with normal renal function.

Patients with hepatic impairment (see *Posology and method of administration*)

In patients with impaired liver function, the metabolism of ramipril to ramiprilat was delayed, due to diminished activity of hepatic esterases, and plasma ramipril levels in these patients were increased. Peak concentrations of ramiprilat in these patients, however, are not different from those seen in subjects with normal hepatic function.

Lactation

A single oral dose of ramipril produced an undetectable level of ramipril and its metabolite in breast milk. However the effect of multiple doses is not known.

Preclinical safety data

Oral administration of ramipril has been found to be devoid of acute toxicity in rodents and dogs. Studies involving chronic oral administration have been conducted in rats, dogs and monkeys. Indications of plasma electrolyte shifts and changes in blood picture have been found in the 3 species.

As an expression of the pharmacodynamic activity of ramipril, pronounced enlargement of the juxtaglomerular apparatus has been noted in the dog and monkey from daily doses of 250mg/kg/d.

Rats, dogs and monkeys tolerated daily doses of 2, 2.5 and 8mg/kg/d respectively without harmful effects.

Reproduction toxicology studies in the rat, rabbit and monkey did not disclose any teratogenic properties.

Fertility was not impaired either in male or in female rats. The administration of ramipril to female rats during the fetal period and lactation produced irreversible renal damage (dilatation of the renal pelvis) in the offspring at daily doses of 50mg/kg body weight or higher.

Extensive mutagenicity testing using several test systems has yielded no indication that ramipril possesses mutagenic or genotoxic properties.

Irreversible kidney damage has been observed in very young rats given a single dose of ramipril.

PHARMACEUTICAL PARTICULARS

List of excipients

Hypromellose, pregelatinized starch (maize starch), microcrystalline cellulose and sodium stearyl fumarate (Ph.Eur.), yellow ferric oxide (E 172, only in Tritace 2.5 mg tablets), red ferric oxide (E 172, only in Tritace 5 mg tablets).

Storage conditions

Store below 30°C

Shelf life

As indicate on the outer box

Nature and contents of container

Tritace tablets are packaged in blister packs containing 28 tablets in PVC/Aluminium strips.

MANUFACTURER

SANOFI S.r.l.

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DATE OF REVISION

March 2022 (Base on CCDS 18/SmPC 16 February 2018)